

1. Course Title	Wavelet Signal Processing
2. Course Instructor	-----
3. Credit Structure	Lecture hours per week: 3 hours Tutorial hours per week: Nil Practical hours per week: Nil Total Credits: 3
4. Course Code	CT423 (3-0-0-3)
5. Program/Semester	B.Tech. (ICT/CS/MNC/EVD) Sem (VI) Technical Elective MTech (ICT with specialization in Machine Learning) Open to Ph.D.
6. Category	Technical Elective
7. Prerequisite courses	Signals and Systems or Introductory Knowledge of Calculus
8. Foundation for	Communication and Signal Processing, Machine Learning, Applied Mathematics Algorithms, EVD.
9. Abstract Content	After a few minutes in a restaurant we cease to notice the annoying sound of surrounding conversations, but a <i>sudden silence</i> reminds us of the presence of neighbors. Our attention is clearly attracted by transients and movements as opposed to stationary stimuli, which we beautifully ignore in our auditory perception. Concentrating on <i>transients</i> is probably a strategy for selecting important information from the overwhelming amount of data recorded by our senses. This course will prepare students in wavelet signal processing algorithms such as signal (image) compression, signal denoising, edge detection, etc.
Suggested Text book(s)	S. Mallat, <i>A Wavelet Tour of Signal Processing</i>. 2nd or 3rd Edition, Academic Press, 2009. P. P. Vaidyanathan, <i>Multirate Systems and Filterbanks</i>

	E. Kreyzig, Introductory Functional Analysis and Applications. DIMENSIONALITY REDUCTION METHODS, SPECIAL SECTION, IEEE Signal Processing Magazine, Vol. 28, Issue 2, 2011. https://ieeexplore.ieee.org/xpl/tocresult.jsp?isnumber=5714377 Deep Learning in Wavelet-Domain - https://arxiv.org/pdf/1811.06115.pdf
Detailed Course Contents	
Topic Name	Content (2 -3 lines per 4 – 6 lectures)
	1. Fourier Analysis (Kingdom): Basic review of Signals and Systems and Related Calculus: LTI Operator (System) and Impulse Response, Fourier transform in $L_1(\mathbb{R})$ and its properties, Defining Fourier Transform in $L_2(\mathbb{R})$, via Density Extension, Global regularity of signal and spectrum decay, 2-D Fourier transform and its properties. Gibbs Oscillations. 2. Sampling and Reconstruction, Shannon's three-step sampling paradigms and its limitations, Removal of Aliasing, Generalized Sampling Theorem in Hilbert Space Framework, Compressive sensing. 3. Wavelet vs. STFT Frames and Applications: Concept of frames (linearly dependent vectors) and application to noise reduction, Design of Sigma-Delta Analog to Digital Converters, Communications, etc. 4. DSP Hardware Architectures: Radix-2 FFT, and DWT. 5. Review of Basic Linear Algebra and Functional Analysis: Metric Space, Vector Space, Linear Combination, Basis, Frames, Normed Space, Banach Space, Inner Product Space, Hilbert Space Introduction to Hilbert spaces, L_2 norm, inner product, $L_2(\mathbb{R})$. 6. Time-Frequency analysis and wavelet: Spectrogram vs. Scalograms, Wavelets: CWT, Admissibility condition, real vs. analytic wavelets, Application of Wavelets to Instantaneous Frequency (IF) Estimation 7. Wavelet Zoom: Lipschitz Uniform vs. Pointwise Regularity, Vanishing Moments of Wavelets, Detection of Singularities, Multiscale Edge Detection, Multifractals,

	Fractal Dimension, Singularity Spectrum, Modeling of Hydrodynamic Turbulence. 8. Multirate DSP: Upsampling, Downsampling, Quadrature Mirror Filterbank (QMF), Paraunitary Filterbanks, Wavelets and Filterbanks. 9. Feature Vector Dimensionality Reduction for Machine Learning- Binning, Log-transform, Principal Components Analysis (PCA), Eigenvalue decomposition, Discrete Cosine Transform (DCT), DWT. 10. Wavelets in Machine Learning: Wav-KAN: Wavelet Kolmogorov-Arnold Networks, Wavelet Scattering , Designing Learnable Wavelet Filters, etc
Course Evaluation Policy (Tentative)	First In-Sem= 15 % Second-In-Sem=15 % Course Project work =20 % Endsem = 40 % Attendance= 10 %
Program Outcomes and Objectives	1) The course builds foundation for the signal processing algorithms, such as short-time Fourier transform (STFT), Gabor transform and wavelet transforms (ie., CWT and DWT). 2) In addition, it will also discuss recently developed auditory transform and it's applications in speech processing. 3) In course evaluation scheme, project work is also considered. Students will be asked to choose any one of the project topic and develop algorithms to implement the objectives of project. This will help students to apply and appreciate the theoretical concepts discussed in the class to real practical problems of engineering and technological interest. 4) One of the important outcomes of this course will be to equip skills to read and understand research papers related to the area of wavelets and time-frequency analysis. This will be quite useful to the students who would like to go for higher studies in near future. 5) This course will be of great value for those students who would like to do their thesis or project work (e.g., BTP, masters, doctoral) in various image and speech related pattern recognition, machine intelligence and signal coding applications.